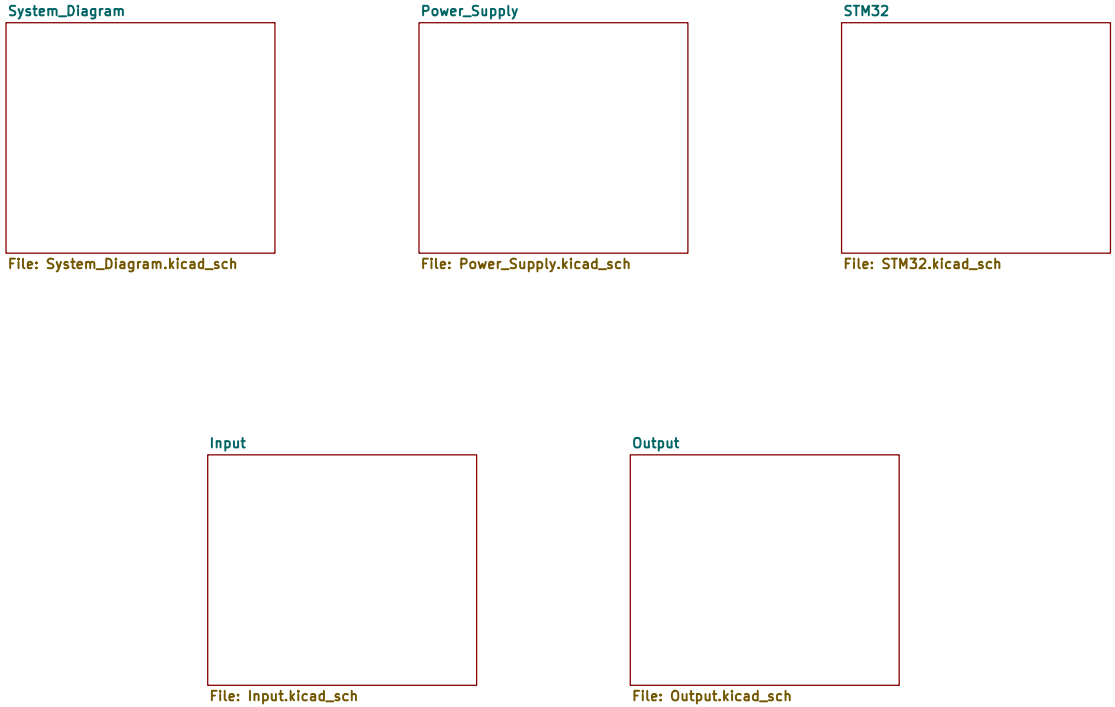
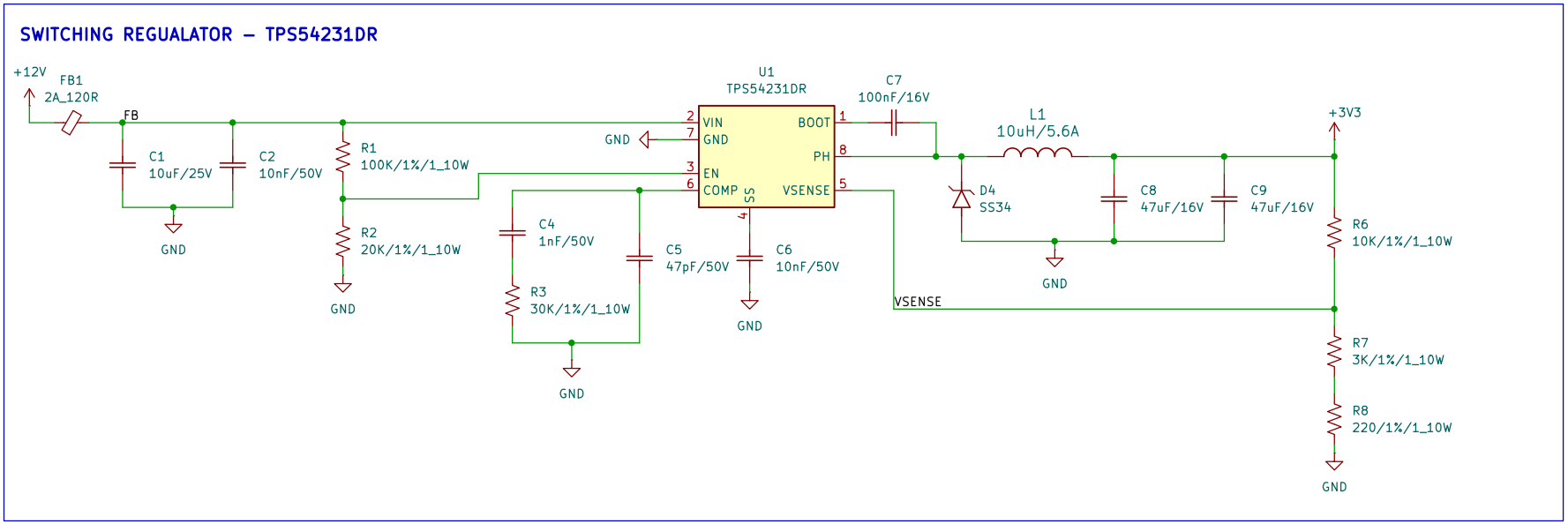
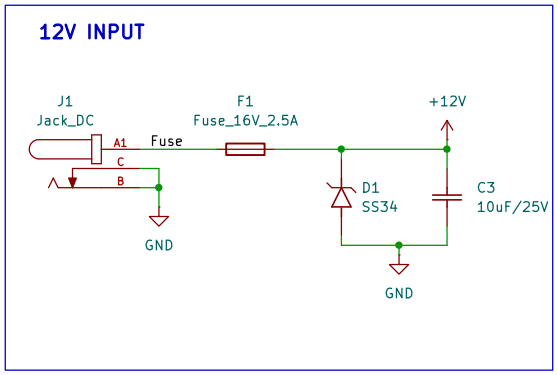
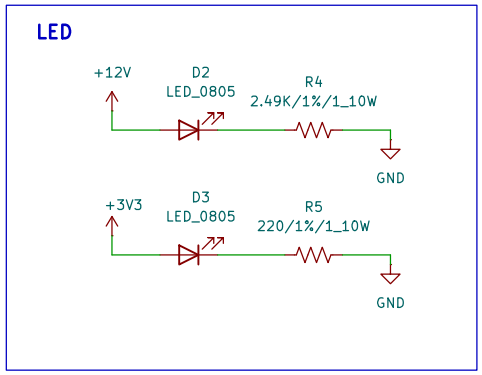
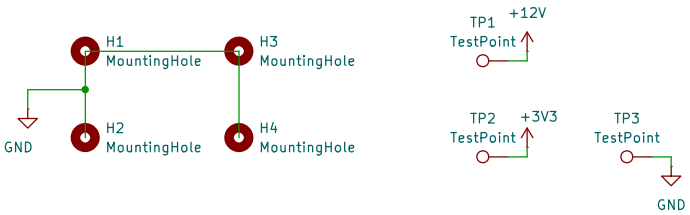
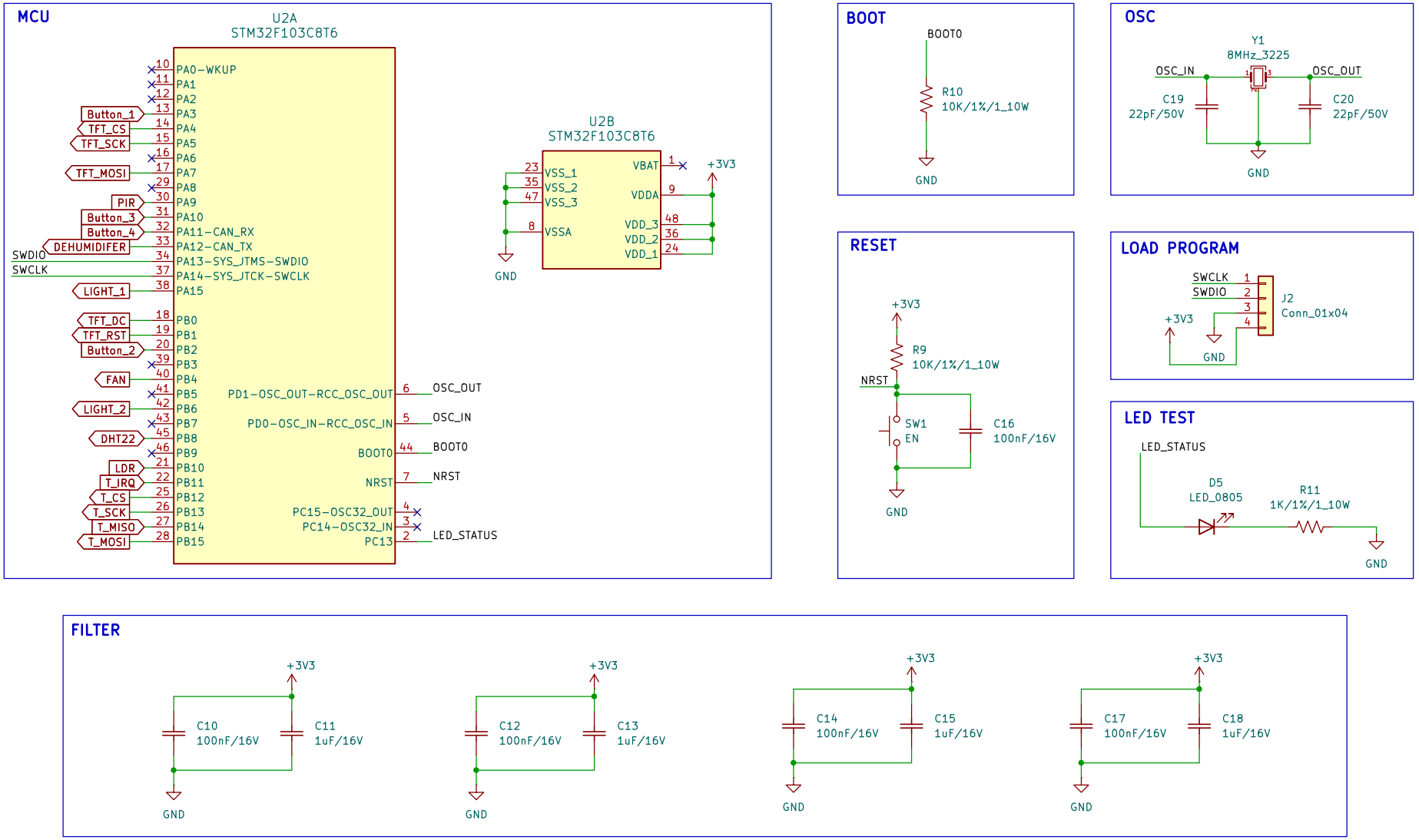

System_Diagram
File: System_Diagram.kicad_sch
Power_Supply
File: Power_Supply.kicad_sch
STM32
File: STM32.kicad_sch
Input
File: Input.kicad_sch
Output
File: Output.kicad_sch

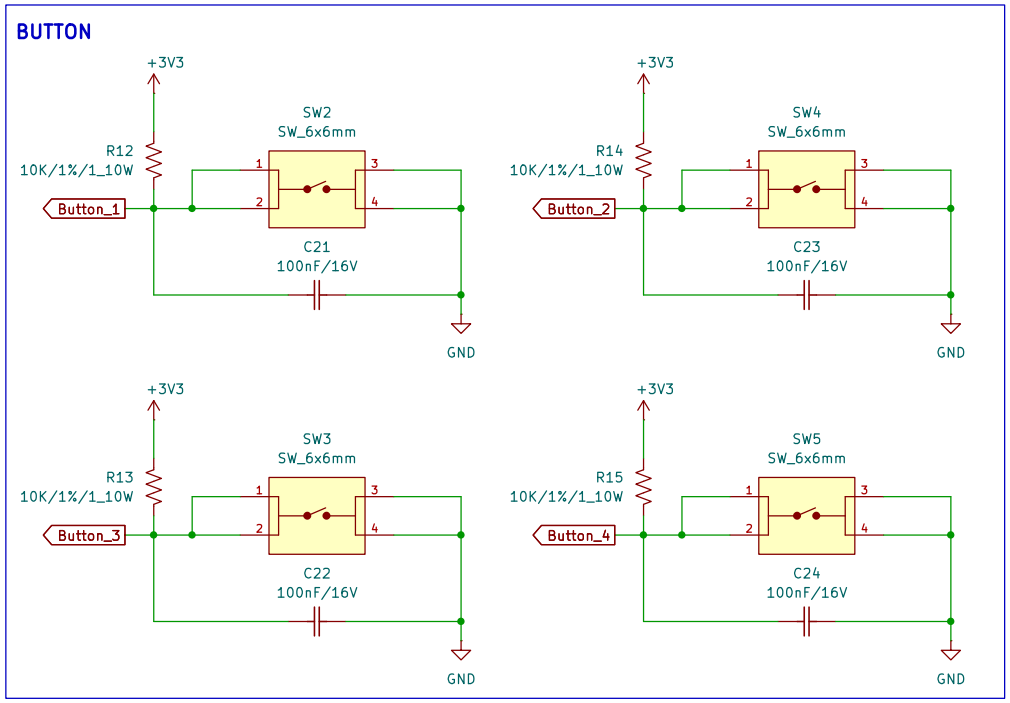

SWITCHING REGULATOR – TPS54231DR
+12V
FB1
2A_120R
FB
C1
10uF/25V
C2
10nF/50V
GND
R1
100K/1%/1_10W
R2
20K/1%/1_10W
GND
U1
TPS54231DR
VIN
GND
EN
COMP
BOOT
PH
VSENSE
SS
GND
C4
1nF/50V
R3
30K/1%/1_10W
GND
C5
47pF/50V
C6
10nF/50V
GND
C7
100nF/16V
L1
10uH/5.6A
D4
SS34
GND
C8
47uF/16V
C9
47uF/16V
+3V3
R6
10K/1%/1_10W
VSENSE
R7
3K/1%/1_10W
R8
220/1%/1_10W
GND

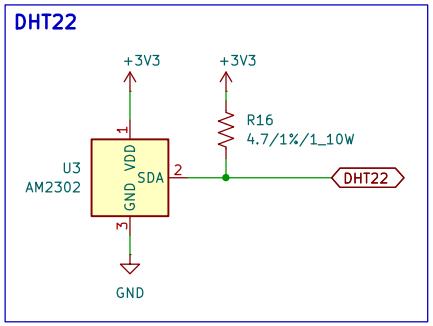

12V INPUT
J1
Jack_DC
A1
C
B
Fuse
F1
Fuse_16V_2.5A
GND
+12V
D1
SS34
C3
10uF/25V
GND

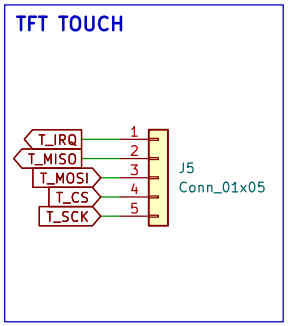

LED
+12V
D2
LED_0805
R4
2.49K/1%/1_10W
GND
+3V3
D3
LED_0805
R5
220/1%/1_10W
GND

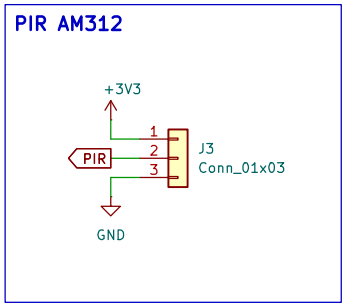

H1
MountingHole
H3
MountingHole
GND
H2
MountingHole
H4
MountingHole
TP1
TestPoint
+12V
TP2
TestPoint
+3V3
TP3
TestPoint
GND

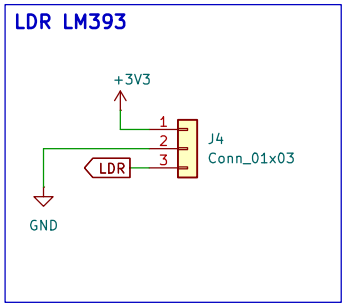

MCU
U2A
STM32F103C8T6
PA0-WKUP
PA1
PA2
PA3
PA4
PA5
PA6
PA7
PA8
PA9
PA10
PA11-CAN_RX
PA12-CAN_TX
PA13-SYS_JTMS-SWDIO
PA14-SYS_JTCK-SWCLK
PA15
PB0
PB1
PB2
PB3
PB4
PB5
PB6
PB7
PB8
PB9
PB10
PB11
PB12
PB13
PB14
PB15
Button_1
TFT_CS
TFT_SCK
TFT_MOSI
PIR
Button_3
Button_4
DEHUMIDIFER
SWDIO
SWCLK
LIGHT_1
TFT_DC
TFT_RST
Button_2
FAN
LIGHT_2
DHT22
LDR
T_IRQ
T_CS
T_SCK
T_MISO
T_MOSI
PD1-OSC_OUT-RCC_OSC_OUT
PD0-OSC_IN-RCC_OSC_IN
BOOT0
NRST
PC15-OSC32_OUT
PC14-OSC32_IN
PC13
OSC_OUT
OSC_IN
BOOT0
NRST
LED_STATUS
U2B
STM32F103C8T6
VSS_1
VSS_2
VSS_3
VSSA
VBAT
VDDA
VDD_3
VDD_2
VDD_1
+3V3
GND
BOOT
BOOT0
R10
10K/1%/1_10W
GND
OSC
Y1
8MHz_3225
OSC_IN
OSC_OUT
C19
22pF/50V
C20
22pF/50V
GND
RESET
+3V3
R9
10K/1%/1_10W
NRST
SW1
EN
C16
100nF/16V
GND
LOAD PROGRAM
SWCLK
SWDIO
+3V3
GND
J2
Conn_01x04
LED TEST
LED_STATUS
D5
LED_0805
R11
1K/1%/1_10W
GND
FILTER
+3V3
C10
100nF/16V
C11
1uF/16V
GND
+3V3
C12
100nF/16V
C13
1uF/16V
GND
+3V3
C14
100nF/16V
C15
1uF/16V
GND
+3V3
C17
100nF/16V
C18
1uF/16V
GND

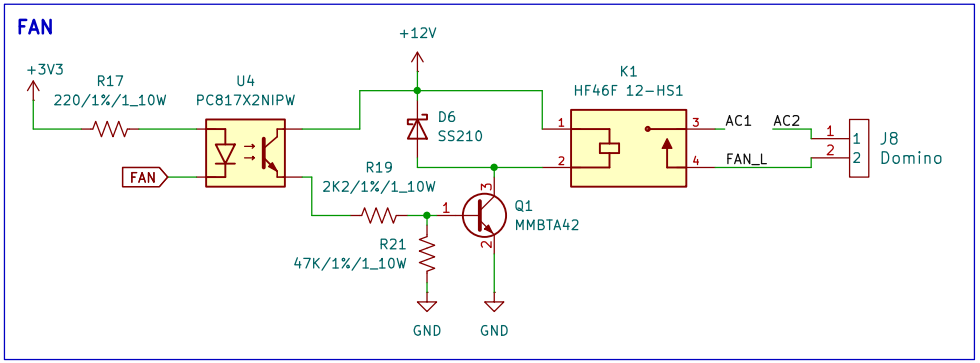

BUTTON
+3V3
R12
10K/1%/1_10W
SW2
SW_6x6mm
Button_1
C21
100nF/16V
GND
+3V3
R14
10K/1%/1_10W
SW4
SW_6x6mm
Button_2
C23
100nF/16V
GND
+3V3
R13
10K/1%/1_10W
SW3
SW_6x6mm
Button_3
C22
100nF/16V
GND
+3V3
R15
10K/1%/1_10W
SW5
SW_6x6mm
Button_4
C24
100nF/16V
GND

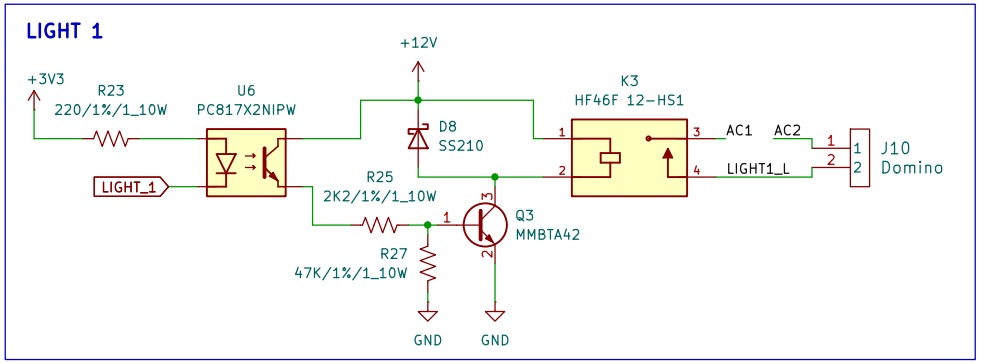

DHT22
+3V3
+3V3
R16
4.7/1%/1_10W
U3
AM2302
GND
VDD
SDA
DHT22
GND

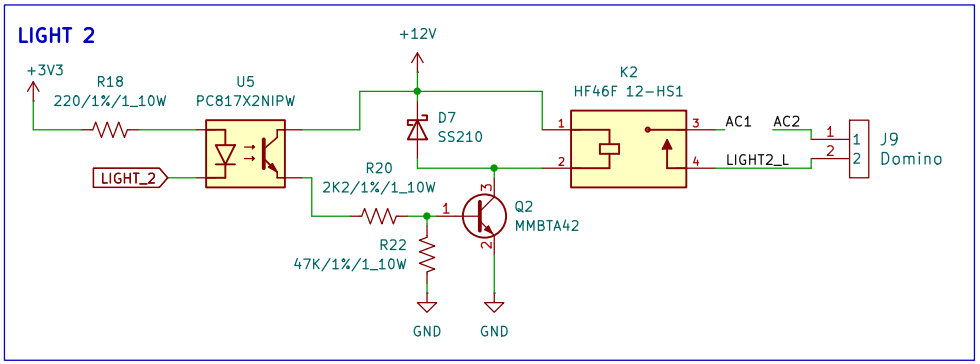

TFT TOUCH
T_IRQ
T_MISO
T_MOSI
T_CS
T_SCK
J5
Conn_01x05

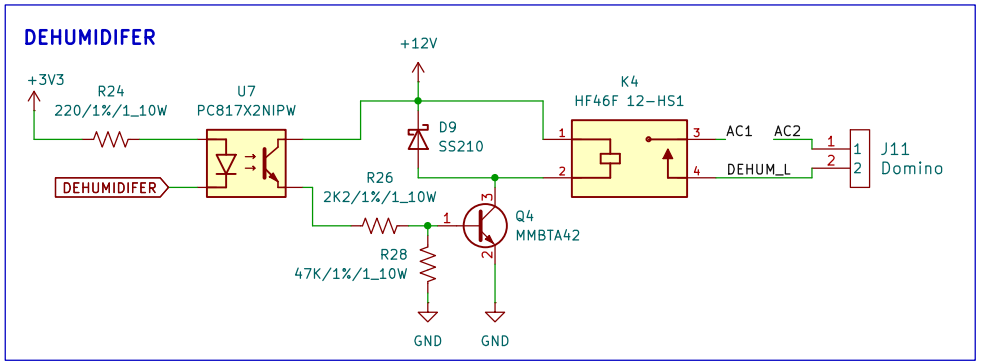

PIR AM312
+3V3
PIR
J3
Conn_01x03
GND

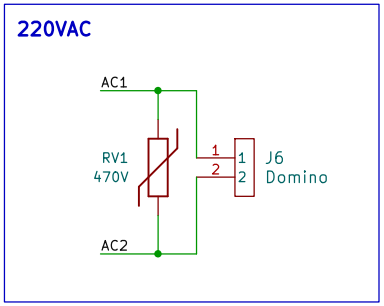

LDR LM393
+3V3
LDR
J4
Conn_01x03
GND

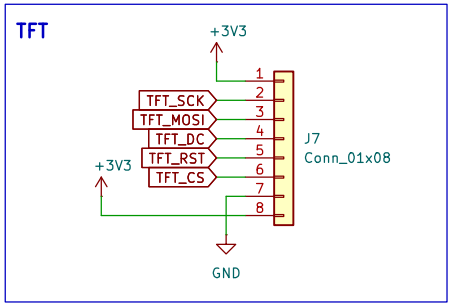

FAN
+3V3
R17
220/1%/1_10W
U4
PC817X2NIPW
+12V
D6
SS210
K1
HF46F 12-HS1
AC1
AC2
J8
Domino
FAN
R19
2K2/1%/1_10W
Q1
MMBTA42
FAN_L
R21
47K/1%/1_10W
GND
GND


LIGHT 1
+3V3
R23
220/1%/1_10W
U6
PC817X2NIPW
+12V
D8
SS210
K3
HF46F 12-HS1
AC1
AC2
J10
Domino
LIGHT_1
R25
2K2/1%/1_10W
Q3
MMBTA42
LIGHT1_L
R27
47K/1%/1_10W
GND
GND


LIGHT 2
+3V3
R18
220/1%/1_10W
U5
PC817X2NIPW
+12V
D7
SS210
K2
HF46F 12-HS1
AC1
AC2
J9
Domino
LIGHT_2
R20
2K2/1%/1_10W
Q2
MMBTA42
LIGHT2_L
R22
47K/1%/1_10W
GND
GND


DEHUMIDIFER
+3V3
R24
220/1%/1_10W
U7
PC817X2NIPW
+12V
D9
SS210
K4
HF46F 12-HS1
AC1
AC2
J11
Domino
DEHUMIDIFER
R26
2K2/1%/1_10W
Q4
MMBTA42
DEHUM_L
R28
47K/1%/1_10W
GND
GND


220VAC
AC1
RV1
470V
J6
Domino
AC2


TFT
+3V3
TFT_SCK
TFT_MOSI
TFT_DC
TFT_RST
TFT_CS
+3V3
J7
Conn_01x08
GND